EFFICIENT FACE TRACK RETRIEVAL COMBINING CONVOLUTIONAL NETS AND GAUSSIAN PROCESSES

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ABSTRACT

Face Track Retrieval from web-scale video databases is one of the important and challenging problems in Computer Vision. In this paper, we present a novel approach to Face Track Retrieval problem that incorporates Spatio-temporal agglomerative clustering, Convolutional Networks (CNNs) and Gaussian Process Latent Variable Model (GPLVM). The key differentiation for our approach comes from exploitation of video-temporal dynamics, usage of very-deep neural networks for face descriptor computation, effective unseen class rejection and usage of Gaussian Processes for automatic feature selection and classification. We finally demonstrate that our approach achieves state-of-the-art performance on six episodes from season 1 of The Big Bang Theory.

Index Terms— *Agglomerative Clustering, Convolutional Neural Nets, Gaussian Processes*

1. INTRODUCTION

Video based face recognition forms the basis for track-retrieval applications and is one of the most desirable and challenging problems in computer vision. Desirability of such applications, arise from practical problems like retrieving shots in a video containing a particular Identity of interest, video abstraction/summarization, semantic video search, large scale face retrieval, personal albums etc.

The challenges for the present problem arise from multiple variations in face pose, illumination, ineffective face detection and recognition approaches. Even though face detection systems like Viola-Jones [1] have shown impressive results on faces with a frontal-view, they fail to achieve consistent performance, when the faces are non-frontal and have a high false-positive ratio. Recently, Trained Deformable Parts Models (DPM) [2] has shown considerable performance improvements in multi-pose face detection. On the other hand, the efficacy of any face recognition system depends mainly on important elements like, robust feature descriptors and effective classification models. Today, Convolutional Neural Networks [3], [4] have given exponential performance boost in large-scale face recognition challenges, by creating robust face descriptors. In addition to that, Gaussian Processes have proven to be, really effective in cases where the input feature spaces are very high dimensional and the number of training examples are limited [5].

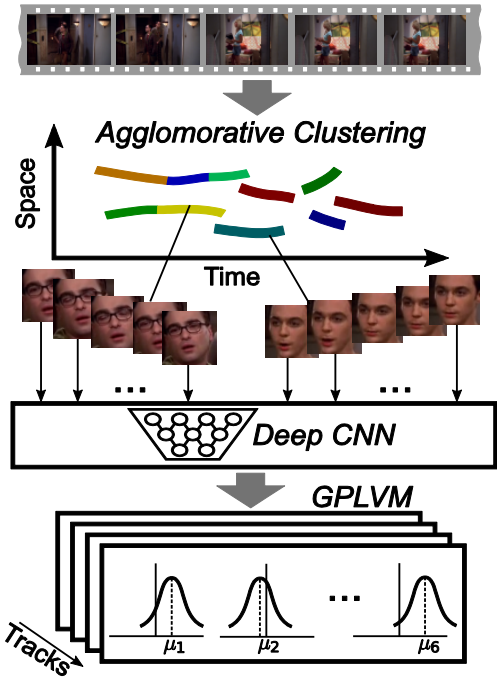


Figure.1 Proposed Face-Track-Retrieval Framework

Contribution

In this paper, we presented a hybrid framework, which combines the effective advantages of, (i) Trained DPM's for robust multi-view face detection (sec 3.1), (ii) Pixel level feature independent Spatio-Temporal Clustering, to create Face Tracks (sec 3.2), (iii) Convolutional Neural Networks for extracting robust Face Track descriptors (sec 3.3) and (iv) Gaussian Processes Latent Variable Model for manifold learning, automatic feature selection, and classification (sec 3.4). Finally, we show that our approach achieved state-of-the-art results on six episodes of season 1, of "The Big Bang Theory". To our best knowledge, the combination of above mentioned approaches have never been used for Face Track Retrieval Problem.

2. RELATED WORK

Our literature survey has been organized to support each of the elements used in our Face Track Retrieval framework. We present the state-of-the-art approaches available and defend our hybrid-framework, in retrospect.

2.1 Creating Face Tracks and Labeling

Once the faces are detected, the next step is to cluster them into tracks (a set of contiguous frames belonging to a particular identity).

Sivic et al. [8] used Kanade-Lucas-Tomasi feature tracker to find initial connections (feature point tracks) between the faces detected. To improve the accuracy of the tracking, the tracker is seeded with features on every face detected and the tracking occurs only in between the detections. Once a set of feature point tracks are obtained, the support for connecting two faces is measured by an overlap score, which is the ratio of number features in common to the number of feature tracks which pass through either of the faces but not both. Any pair of faces, which have an overlap score of 0.5 or more, are considered for connection. To reduce the total number of faces in each track, agglomerative clustering is used with no pose bias.

Given a set of detected faces, Ortiz et al [7] used a combination of location metric and an appearance metric to cluster them into tracks. Assume (x_1, y_1, w_1, h_1) and (x_2, y_2, w_2, h_2) are the x, y, width and height of the previous and present face-bounding box detected. The criteria for merging a new face into an existing track, is decided by (i) a spatial distance, which is the ratio of maximum sized bounding box encompassing both the detections to the size of larger bounding box, (ii) an appearance distance, which is the ratio of RGB histograms of the last face in the track to the new face detected. A new face will be merged into an existing track if the location and appearance metrics are similar, if not, a new track is created. Ramanan et al. [9] used a combination of color histograms for hair, face and torso to group a set of detected faces into tracks.

See et al. [10] used Spatio-Temporal Hierarchical Agglomerative Clustering (STHAC) and demonstrated the superiority of this approach on UCSD Honda Video-Face-Recognition dataset.

Once the Face Tracks are created, the next step is to annotate them for training and testing. Everingham et al. [11] used a combination of subtitles and scripts to weakly label the Face Tracks. The subtitles and the timestamp are extracted using an openly available software called SubRip which uses OCR. The extracted subtitles are further corrected using a spelling correction algorithm, to avoid the mistakes due to OCR. Scripts for each video are aligned with the subtitles to find who is speaking and when. This kind of annotation can propose one or more possible speaker names for each frame, to clear this ambiguity, Speaking Face Detection, which classifies a give face as speaking versus non-speaking using significant lip-motion is used. In addition to this approach, Bauml et al. [6] also incorporated the entropy loss of unlabeled data and temporal constraints like "two tracks which overlap temporally cannot belong to the same person" to improve the annotation accuracy. Ortiz et al [7] relied on web-scraping and labeled datasets like LFW and PubFig to create a large dictionary of still-faces, of few hundred people for training.

2.2 Feature Extraction and Classification

A combination of different feature descriptors like Harris features, Maximally Stable External Regions (MSER), Histogram of Oriented Gradients (HOG) and Local Binary Pattern Histograms (LBPH), Gabor Wavelets [7], SIFT [8] with SVM, Logistic Regression, Bayesian Nets and Mean Sequence Sparse Representation based classification schemes [7] have been extensively used in video based face recognition systems. Simonyan et al [4], has shown exponential performance improvements on large-scale object/face recognition challenges using Convolutional Neural Networks. Chan et al. [12] used a combination of PCANet based features with Linear SVM and achieved results comparable to state-of-the-art on large-scale object & face recognition challenges. Gaussian Processes for classification are also proven to be effective, especially for face verification/recognition approaches, where training data is limited and feature descriptors are very high dimensional and sparse [13].

The main limitations of the video-based face recognition approaches can be summarized as follows (i) *Usage of Rigid-Template based approaches for face detection*: Zafeiriou et al [14] have proved that, face detection systems, based on Deformable Parts based Models are more efficient in terms of annotation effort required, amount of training data, training & test time performances and are more robust to unseen face views in comparison to rigid template based models. In our present framework, we used a pre-trained Deformable Parts based Model proposed in [2] as our face detector. (ii) *Usage of feature based tracking to create Face Tracks*: Feature based tracking approaches are very sensitive to background clutter, occlusion, illumination, and wide pose variations. In our present framework we use agglomerative clustering, exploiting the Spatio-Temporal and further improve the accuracy using pixel-level information. (iii) *Relying on script and subtitle alignment based approaches for labeling Face Tracks*: In most of the cases, the scripts are not at all available. Considering scripts are available, extracting subtitles using software's like SubRip and aligning them with scripts is error-prone and computationally costly, (iv) *Usage of in-effective face descriptors and Classification schemes which would require*

extensive cross validation: In our proposed framework, we used state-of-the-art Deep Neural Networks for face descriptor extraction and Gaussian Processes as our non-parametric classification model.

3. OVERVIEW OF OUR APPROACH

3.1. Face Detection

The Head Hunter (face) detector [2] used in the present framework has a true-positive rate of 87.1% on FDDB, achieving a state-of-the-art performance. In addition to face detection, this method also provides us with the orientation information of the face, belonging to one of the six poses (1: right profile, 2: left profile, 3: right half profile, 4: left half-profile, 5-6 frontal faces). This information is further utilized to flip right-half profiles faces, to improve the recognition accuracy.

3.2. Clustering and Pruning

Our main objective in this paper is to retrieve the Face Tracks of the actors. The initial step for the retrieval is to, index/cluster the video database into groups of tracks. In order to achieve this, we used Hierarchical Clustering on the face crops detected. Hierarchical Clustering methods can be divided into either top-down or bottom-up approaches. The Hierarchical Clustering [10] used in this paper is a bottom up approach, and is explained below in detail.

a. Every face crop is treated as a singleton cluster (self-similarity property). We represent each face crop as $[x_1, y_1, x_2, y_2, \alpha t]$ where (x_1, y_1) are the normalized top left corner, (x_2, y_2) being the bottom right corner, 'α'is a coefficient which adjusts intra-cluster temporal-variance relative to the spatial-variance and t is the frame timestamp.
b. Choice of distance metric: In this paper we chose the most widely used Ward's minimum variance method with Euclidean distance.
c. Once the convergence is achieved, we cut the dendrogram into an appropriate number of clusters chosen empirically, analyzing the track-wise false-positive ratio which is widely affected by the choice of large/small number Face Track clusters.

It is evident from our clustering approach that, we only exploited the spatial and temporal information and completely ignored pixel level information. Our argument is that, the likelihood of false-positives due to such an approach is extremely minimal. To avoid the minutest uncertainty of having false-positives (*for instance, Penny's face in Leonard's track)* due to not considering pixel level information, we further prune (cluster and remove outliers) each clustered Face Track using color histogram as our feature descriptor and Chi-squared distance as a similarity metric. We use complete-linkage clustering for this process, as it is based on farthest distance it is more efficient in detecting outliers. We also argue that, considering pixel level information for the initial clustering would be redundant, computationally expensive and in-efficient. For each actor, from the extracted track clusters we chose 3 Face Tracks with frontal profiles (as explained in section 3.1) as our training exemplars and the total number of clusters for each episode is fixed at 500.

3.3. Feature Extraction

In this paper, we used a pre-trained 39-layered Deep Neural Network proposed in [3]. This network achieved state-of-the-art results on large-scale face recognition/verification datasets like Labeled Faces in the Wild, YouTube Faces etc. For our experiments, we froze the last six layers of the network and used it as a fixed feature extractor, which would give us a 4096 dimensional feature descriptor for each face.

3.4. Training and Classification

In this paper, we used Gaussian Process Latent Variable model proposed by Rasmussen et al. [15]. The main motivation for using Gaussian Processes comes from the non-parametric nature of the formulation, where the hyper-parameters can be learned completely from the data without an explicit need for cross validation. In addition to that, the latent variable formulation (section 3.4.1) offers an automatic way of feature dimensionality reduction, thereby improving the recognition accuracy. We also explored automatic feature selection using ARD (section 3.4.2) as mentioned in [17]. We trained six (number of actors considered) binary classifiers, using three track exemplars per each class and RBF-kernel with two hyper parameters (support width and output variance) for our covariance matrix (eq. (1))

$$K_{n,m}(x_n, x_m) = \alpha \ \exp\left(-\frac{\gamma}{2}(x_n - x_m)^T (x_n - x_m)\right)$$

With Hyper-parameters $\theta = (\alpha, \lambda)$ (1)

3.4.1 Gaussian Process Latent Variable Model

Given a very high-dimensional input feature space $X = [x_1, x_2, ...x_N]^T \in R^D$, the objective of GPLVM is to map it to a low dimensional latent space $Z = [z_1, z_2, ...z_N]^T \in R^d$ where $d = D$ [16]. Given a covariance matrix K, the locale of such low-dimensional latent points can be obtained by maximizing the Gaussian Process likelihood shown in equation (2) over Z.

$$p(X \mid Z,\theta) = \frac{1}{\sqrt{(2\pi)^{ND} \mid K \mid^{D}}} \exp\left(-\frac{1}{2} tr(K^{-1}XX^{T})\right) \quad (2)$$

Once the Gaussian Process likelihood for the latent space is calculated the posterior distribution of our latent space can be estimated using a simple equation below

$$p(Z,\theta \mid X) = \frac{1}{z_a} p(X \mid Z,\theta) p(Z) p(\theta) \quad (3)$$

Optimal latent function '$\hat{z}$' which maps the given high dimensional feature set into a low dimensional latent space can be estimated by maximizing the posterior distribution above.

3.4.2 Automatic Feature selection using ARD

For instance if the covariance matrix of the data under consideration can be estimated as shown in equation (4)

$$K_{nm} = \alpha \exp\left[-\frac{\gamma}{2}\sum_{d=1}^{D}\left(x_n^{(d)} - x_m^{(d)}\right)^2\right] where\ \gamma = \frac{1}{r_d^2} \quad (4)$$

We can clearly observe that as 'r_d', which is the scale of the function along the dimension d, tends to infinity ($r_d \rightarrow \infty$), the function would vary less as a function of $x^{(d)}$, and the d'th feature dimension would become completely irrelevant. It shows that by being able to learn the length scales, we can clearly accomplish automatic feature selection.

3.4.3 Classification

Given a new test face, we first compute the feature representation using the method mentioned in sec 3.3. And then we calculate the latent representation z_* using the mapping learnt (sec 3.4.1, 3.4.2). Once the latent representation is calculated and a covariance matrix (K) is learnt from the training data & labels, the probability can be estimated using a simple sigmoid function (eq. 5) as shown below.

$$\pi_* = \Theta\left(\frac{f_*(z_*)}{\sqrt{1+\sigma^2(z_*)}}\right) \quad (5)$$

$$where\ \ \sigma^2(z_*) = K_{**} - K_{*N}(K_N + \sigma^2 I)^{-1} K_{N*} + \sigma^2\ \ and$$

$$f_*(z_*) = K_* K^{-1} \hat{z}$$

All the Face Tracks except the ones with completely non-frontal face images are classified for our experiment.

4. EXPERIMENTS

In this section, we evaluate the capability of our proposed Face Track retrieval framework from two points of view: 1. Face recognition; 2. Reject the tracks of the characters that are not present in the training set.

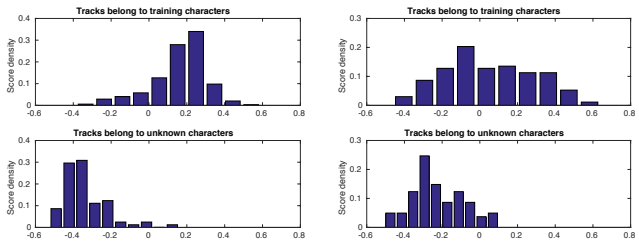
4.1 Experimental Setup

We tested our Face Track retrieval framework on the all the six episodes of season 1 of The Big Bang Theory TV series. The show is primarily centered on six characters. Each episode is 20 minutes long, with mostly full view shots (face & body), which contain multiple people. This TV series has been already been used in [6] for semi supervised track labeling and speaker detection. Unfortunately, the provided ground truth in [6] is based on PAL (European display system) while we only have access to the NTSC version of DVDs (American standard).

In our framework, first we detect faces in all episodes, and then the faces in each episode are clustered into 500 tracks. We further prune each Face Track to remove the possibility of wrongly clustered faces as explained in sec 3.2. To study how much our system is able to reject the unknown characters, we only consider the six main characters as our training set. Therefore we train our GPLVM as a 6-class problem (one versus all), with three track exemplars per each class. Table (1) describes the attributes like total number of detected faces for each episode, number of unknown tracks and number of tracks which contain at least one false-positive (for instance, Penny's face in Leonard's track).

The task here is to label each track with the identity of the person; additionally we require Face Tracks belonging to characters not in the training set to be identified as 'unknown'. In each track, faces are ordered based on the classifier confidence score and the track label is chosen using the majority voting among k top faces. By doing this, we are eliminating the effect of outlier faces, such as those with severe occlusion, illumination, pose variation, and motion blur (very frequent in video face recognition).

Baseline: We compare our approach with PCANet feature descriptor [12], which has achieved state of art results on many face recognition datasets such as Extended Yale B, FERET, and AR dataset. This network has been trained on MultiPIE dataset and the architecture has been provided in [12]. Additionally, we use linear SVM for classification.



(a) (b)

Fig. 2. Score density function for correctly classified tracks and unknown characters. (a) CNN features and GPLVM classification (our approach). (b) Baseline: PCANet features and SVM classifier.

4.2 Unknown Face Rejection

In open universe face recognition settings, the objective is not just to associate a given test face with one of the training labels, instead there is a probability measure which defines whether a given test face belongs to one of training classes or if it is an unseen/unknown class. In this paper we used a frequentist inference to estimate whether a given test face is unknown using the score-density function of correctly classified tracks versus the score-density function of the unknown tracks. Figure 2, gives the empirical validation for our approach. In Figure 2, the histogram of the classifier scores for two different Face Track categories have been plotted: the top histogram belongs to the known characters (the ones who are present in the training set) and bottom belongs to the unknown tracks. Figure 2(a) shows our proposed approach, where there is a clear delineation between the score histograms of the known and the unknown tracks, as opposed to Figure 2(b), which is the baseline. This result is corroborated by the receiver operating characteristic (ROC) curve of the $p(unknown)$, shown in Figure 3, in which an equal error rate (EER) of 8.3% and 28% is achieved for our approach and baseline respectively.

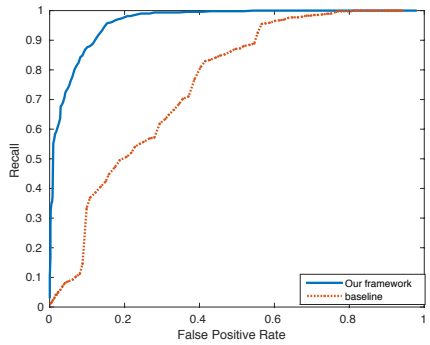


Fig. 3. ROC curve of probability of unknown.

4.3 Face Recognition

The results of our Face Track recognition, tested on six episodes of The Big Bang Theory are shown in Table (2). In the first scenario, we show our results without any unknown rejection, where each track is assigned to the closest main character. The first two columns of Table 2 show the results of our approach in comparison to our baseline. It can be observed that, our approach had a consistent improvement in accuracy on all episodes, with an average improvement of 20.87%, when compared to our baseline.

In the second scenario, we first evaluate whether a test Face Track should be labeled as unknown, using our unknown rejection (Sec. 4.2), if not, we classify the Face Track using our trained model. The accuracies of our approach with unknown rejection are shown in Column 4 of Table 2. We observed two important things, (i) the average accuracy increased by 3.58%, when unknown rejection is considered. This improvement is especially evident when the number of unknown tracks is high (for instance, episode 5), (ii) the unknown rejection didn't improve the accuracies, in the baseline approach and we account this, to the fact that PCANet features are not as discriminative as CNN features (also validated in sec 4.2). Furthermore, we also investigated the reason for accuracy drop, in episode 6 and found that most of the miss-classified tracks belong to the characters having a Halloween face makeup (an example has been shown in Figure 4).

Table 1. Detailed attributes of the dataset

	Total number of faces	Number of Unknown Tracks	Number of tracks with at least one false-positive
Episode 1	14412	6	76
Episode 2	10566	1	14
Episode 3	11352	61	8
Episode 4	7873	87	5
Episode 5	11674	84	16
Episode 6	10590	89	6

Table 2. Face Track recognition accuracy

	No Unknown rejection		Unknown Rejection	
	PCANet +SVM (%)	**Our method(%)**	**PCANet +SVM (%)**	**Our method(%)**
Episode 1	87.90	97.45	84.24	98.17
Episode 2	67.03	97.16	63.50	95.41
Episode 3	75.60	87.69	76.70	92.31
Episode 4	58.79	82.21	57.48	86.12
Episode 5	60.87	79.41	63.39	92.22
Episode 6	58.28	89.80	59.41	90.93
Average	**68.08**	**88.95**	**67.45**	**92.53**

Fig. 4. (a) Selected faces from **training** track. (b) Selected faces from **correctly** classified test track (c) selected faces from **incorrectly** classified test track.

5. CONCLUSION

In this paper we present an efficient Face Track retrieval framework, which exploits/combines (i) Pixel level feature independent Spatio-Temporal Clustering for creating face tracks, (ii) deep neural networks for feature extraction, (iii) effective unseen class rejection and (iv) Gaussian Processes for classification. Our proposed method is tested on six episodes of The Big Bang Theory, with six important characters and achieved a state-of-the-art average accuracy of 92.53% with most minimal supervision for the training data annotation, hyper-parameter tuning and inference.

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